NHS A&E Project — Data Quality Log

Nov 1, 2026 · @emeka

Scope

Ten data quality issues were found while cleaning 24 months of NHS England A&E data. All ten are resolved in the Python pipeline, and the cleaned output reconciles exactly to the published England totals for October 2024.

Item	Detail
Dataset	Monthly A&E Attendances and Emergency Admissions (MSitAE), NHS England
Level	Provider (one row per organisation per month)
Period	October 2024 to September 2026 (24 monthly CSV files)
Raw input	`data/raw/ae_YYYY_MM.csv`, 22 columns, 192 to 198 organisations a month
Cleaned output	`data/cleaned/ae_clean_2024_10_to_2026_09.csv`, 4,721 rows x 25 columns
Pipeline	`clean_ae(filepath)` in `notebooks/01_data_exploration.ipynb`, looped over all files and combined with `pd.concat`
Revised files	October and November 2024 were downloaded as the revised releases (15 May 2025)

The pipeline applies seven steps to every file: load, standardise column names, trim text, remove the England total row, convert `period` to a date, add attendance and breach totals, and flag organisations with a Type 1 department.

Issue log

The two most serious issues were the England total row (DQ-01, DQ-02), which would have doubled every national figure, and two column names that automated cleaning silently changed in meaning (DQ-05, DQ-06).

ID	Issue	How it was found	Impact if left	Fix	Status
DQ-01	Every file contains an England total row (`TOTAL` in Oct 2024)	`describe()` showed a Type 1 maximum of 1,422,317 against a	National totals double-counted; summary	Total row removed in step 4; provider sums	Resolved

ID	Issue	How it was found	Impact if left	Fix	Status
	alongside the providers	median of about 7,450	statistics inflated	reconciled to it (see Validation)	
DQ-02	The total row's label changes between files: `Total` (title case) in Jun 2026	Loop failed on `ae_2026_06.csv` with a date error on the value "Total"	Exact-match filter missed it, so totals would be doubled from that month	Filter made case-insensitive and checks both `org_code` and `period`	Resolved
DQ-03	`Period` stored as text, e.g. `MSitAE-OCTOBER-2024`	`.info()` showed type `object`	No time-series analysis or date table possible	Prefix removed, converted with format `%B-%Y`; dates set to the 1st of the month	Resolved
DQ-04	Column names are long and contain spaces, & and `-`	Column list review	Error-prone code; awkward fields in SQL and Power BI	Automated snake_case clean, then an explicit rename map	Resolved
DQ-05	Source typo `4-12 hs` (for hrs); automated cleaning then removed the hyphen, giving `412_hs`	Review of cleaned names	Column reads as "412 hours"	Renamed `dta_to_admission_4_12hrs`	Resolved
DQ-06	`12+ hrs`: automated cleaning removed the +, giving `12_hrs`	Review of cleaned names	Column reads as exactly 12 hours instead of more than 12	Renamed `dta_to_admission_above_12hrs`	Resolved
DQ-07	Emergency admission names produced double underscores from `-`	Review of cleaned names	Untidy, easy-to-mistype names	Renamed, e.g. `emergency_admissions_via_ae_type_1`	Resolved
DQ-08	"Other emergency admissions" (not via A&E) is easily confused with admissions via Other A&E departments	Review of rename map, where the two were briefly swapped	Wrong measure used in analysis	Named `emergency_admissions_not_via_ae` and `emergency_admissions_via_ae_others`	Resolved

ID	Issue	How it was found	Impact if left	Fix	Status
DQ-09	Files mix major trusts with standalone sites (walk-in centres, UTCs, community services)	Inspection of rows with zero Type 1 attendances	Unfair trust comparisons; standalone sites score near 100%	`has_type1` flag added; Oct 2024: 122 with Type 1, 76 without	Resolved
DQ-10	Number of reporting organisations varies by month (198 in Oct 2024, 192 in Jun 2026)	Row counts per file	Changes in totals may partly reflect reporting changes, not activity	Documented; month-by-month counts to be reviewed	Accepted
DQ-11	18 of 130 trusts with a Type 1 department have fewer than 24 months of data (minimum 1 month), likely from mergers or organisation code changes	Trust ranking returned 130 trusts instead of about 120; COUNT(DISTINCT period) showed partial records in some trusts	Trusts ranked on as little as one month against others ranked on 24; unfair comparison	Trust rankings limited to the 112 trusts with all 24 months using HAVING COUNT(DISTINCT period) = 24 (sql/05_q2_trust_ranking.sql)	Resolved
DQ-12	Croydon Health Services (RJ6) reports 0 twelve-hour waits in Jun 2025, against 441 to 873 in every other month, with normal admissions (1,143)	Monthly check after Croydon topped the 12-hour wait rate ranking (534 per 1,000 admissions). Zero-wait trusts per month stayed at 5 to 10 throughout, so this is specific to Croydon, not a national gap	Croydon's 24-month total is understated by roughly 600 waits, so its rate is slightly underestimated; its top ranking is conservative	Kept as reported and documented; Croydon's result stated with this caveat. Its admissions via A&E (about 1,100 a month) are low for a major acute trust and worth confirming against trust-level sources	Accepted

Validation

Removing the total row lost no data: the 198 provider rows for October 2024 sum exactly to the published England totals. All checks on the combined file passed.

Check	Expected	Result
Oct 2024 total attendances, sum of providers vs removed total row	2,360,536	2,360,536, exact match
Oct 2024 attendances over 4 hours, sum of providers vs removed total row	635,342	635,342, exact match
`clean_ae()` on Oct 2024 reproduces the step-by-step result	(198, 25); 2,360,536; datetime64	All three match
Files found by `glob`	24	24
Combined file built	24 months stacked	4,721 rows x 25 columns
Missing values, Oct 2024	None	None in any column
Oct 2024 4-hour performance (all types) vs published figure	73.0% (NHS England commentary)	73.08%: 0.1 point higher, likely because the file is the May 2025 revision
Distinct months in combined file	24	24, as expected
Total rows left in combined file	0	0, as expected

The 4-hour figure uses (total attendances - attendances over 4 hours) / total attendances, with booked appointments included in both counts.

Method notes

- **Raw files are never edited.** All fixes are in code, so the pipeline can be rerun when a new month is published.
- **Counts in Python, percentages in Power BI.** Only counts are stored. Percentages are DAX measures over summed counts, because averaging row-level percentages would weight a walk-in centre the same as a major trust.
- **"Over 4 hours" counts breaches.** Within-4-hours is derived as total minus breaches. The clock runs from arrival to admission, transfer or discharge.
- **Automated cleaning must be reviewed.** The regex that standardised names also removed a hyphen and a +, changing two columns' meaning (DQ-05, DQ-06).

- **Test on every file, not just the first.** The total-row label change (DQ-02) only appeared when the pipeline ran on the June 2026 file.
- **Text columns are trimmed** of leading and trailing spaces as a precaution; no issue was traced to whitespace.
- **Specialist children's trusts are kept but flagged.** Three of the top five Type 1 performers are children's trusts (Sheffield Children's 91.7%, Alder Hey 83.8%, Birmingham Women's and Children's 79.9%). Their patient mix differs from general acute trusts, so comparisons are not like for like. They remain in the SQL results and will be filterable by trust type in Power BI; the best general acute trusts are Homerton (80.6%) and Calderdale and Huddersfield (80.3%).
- **Pressure bands are a defined method, not NHS categories.** Trusts are banded on Type 1 4-hour performance against the 95% standard: 85-94.9% near standard, 75-84.9% moderate, 60-74.9% high, below 60% severe (sql/08_q2_pressure_bands.sql). NHS England measures trusts against 95% on all A&E types, so these Type 1 results are stricter than the official measure.